



P0 roadmap

Students:

Nancă Cristian
Petriia Iulian-Nicolae
Stan Alexandru-Gabriel

Group: 441Gb

Project chosen - 6 - LTE vs 5G NR



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I. Feasible

We will follow the scenario defaults, as we have not yet encountered any significant hurdles that would force us to pivot. This includes confirmed handover, successful runs of lena-simple-epc and ctte-nr-demo. The given roadmap remains unchanged.

II. Specific

a. Hardware allocation

All experiments are conducted on an isolated virtual machine configured using Oracle VM VirtualBox (v7.1.10 r169112) running Ubuntu 20.04.1 LTS. The VM is allocated 6 virtual cores from a 13th Gen Intel Core™ i7-1355U processor, 9.2 GiB of dedicated RAM, and 26.8 GB of disk space. This allocation conforms directly to the project guideline constraint (8–16 GB RAM laptop) while leaving sufficient host overhead.

a. Simulator version & Build Resolution

We started by reading the documentation for the ns-3, namely ‘<https://www.nsnam.org/docs/installation/html/quick-start.html>’ and we have run into a problem related to the version for the ns-3 from their GitHub (accessed on **06.10.2026** – ‘\$ git clone <https://gitlab.com/nsnam/ns-3-dev.git>’). There was a problem within the main function that did not allow the build in function to be executed directly. An alternative identified was to use directly the version <https://www.nsnam.org/releases/ns-3-48/documentation>.

To avoid build instability identified on the bleeding-edge upstream development branch (ns-3-dev), we aligned our setup with the recommended stable distribution release ns-3.48. NS-3 was configured via CMake (`./ns3 configure --enable-examples --enable-tests`) with runtime asserts and logging enabled. System integrity was verified by running `./test.py`, which yielded 812 of 812 tests passed (0 failed, 0 skipped).

b. 5G-LENA Module Integration

For 5G New Radio support, the CTTC 5G-LENA module was cloned into `contrib/nr` on branch `5g-lena-v5.1.y`, satisfying the Project 6 prerequisite of version ≥ 5.0 required for handover modelling. All critical numerical and linear algebra dependencies—specifically Eigen3 (for MIMO matrix processing) and SQLite—were verified as active during the build configuration.



III. P0 evidence

5G NR Baseline (cttc-nr-demo): Built and executed without errors. Captured UDP flows showed expected performance: Flow 1 (10.24 Mbps offered load) achieved 10.24 Mbps throughput with 0.276 ms mean delay; Flow 2 (102.40 Mbps offered load) achieved 102.23 Mbps throughput with 0.901 ms mean delay (aggregate mean delay: 0.588 ms).

LTE EPC Baseline (lena-simple-epc): Built and executed successfully, validating standard LTE PHY/MAC/RLC logging via the generation of discrete uplink and downlink trace logs (DlMacStats.txt, UlPdcPStats.txt, DlRsrpSnrStats.txt, etc.).

Output of cttc-nr-demo (./ns3 run cttc-nr-demo)

```
Flow 1 (1.0.0.2:49153 -> 7.0.0.2:1234) proto UDP
Tx Packets: 6000
Tx Bytes: 768000
TxOffered: 10.240000 Mbps
Rx Bytes: 767744
Throughput: 10.236587 Mbps
Mean delay: 0.276044 ms
Mean jitter: 0.030032 ms
Rx Packets: 5998
Flow 2 (1.0.0.2:49154 -> 7.0.0.3:1235) proto UDP
Tx Packets: 6000
Tx Bytes: 7680000
TxOffered: 102.400000 Mbps
Rx Bytes: 7667200
Throughput: 102.229333 Mbps
Mean delay: 0.900967 ms
Mean jitter: 0.119907 ms
Rx Packets: 5990
```

```
Mean flow throughput: 56.232960
Mean flow delay: 0.588505
```

Output of lena-simple-epc (./ns3 run lena-simple-epc)

From this scenario test we have identified a couple of files that were generated for this scenario that are listed in the GitHub folder. The primary analysis of these logs shows that for the topology and connectivity, 2 active cells (cellId 1 and 2), each serving one UE (IMSI 1 and 2, RNTI 1), for the channel and physical layer, a pristine radio channel condition ($SINR \cong 1.64 \cdot 10^6$ linear, approx. 62.15dB with negligible interference (1.35×10^{-20} W)),

allowing the scheduler to maintain the highest MCS index ($mcs = 28$) across both Downlink and Uplink without hybrid ARQ retransmissions ($rv = 0$, $ndi = 1$, $correct = 1$). For the transport and data plane (PDCP/RLC), downlink (DL) Uniform delivery of 1,054-byte PDUs at ≈ 3.53 ms delay (RLC delay = 3.00 ms) with 0% packet loss (12 Tx = 12 Rx PDUs per UE over the simulation interval) and for the Uplink (UL), Stable transmission of 1,054-byte PDUs at ≈ 11.93 –12.26 ms delay (RLC delay = 4.71 ms, aggregating two PDCP SDUs into 2,112-byte RLC PDUs).

Layer / Metric	LTE EPC Baseline (lena-simple-epc)	5G NR Baseline (cttc-nr-demo) TXT
Topology	2 eNodeBs, 2 UEs (1 UE / cell)	1 gNodeB, 2 UEs (2 UDP Flows)
Channel Quality	SINR ≈ 62.15 dB, Interf. $\approx 1.35 \times 10^{-20}$ W	Urban Macro default, 3.5 GHz
Modulation / Coding	Max MCS 28 (DL TB: 2,196 B, UL TB: 2,292 B)	Adaptive NR MCS tables
Packet Delivery Ratio (PDR)	100.0% (DL: 24/24 PDUs, UL: 24/24 PDUs)	99.97% (Flow 1), 99.83% (Flow 2)
End-to-End Delay	3.53 ms (DL PDCP), 12.04 ms (UL PDCP)	0.276 ms (Flow 1), 0.901 ms (Flow 2)
Transit / Jitter	RLC Delay: 3.00 ms (DL), 4.71 ms (UL)	Mean Jitter: 0.030 ms (F1), 0.120 ms (F2)

IV. Risks with fallbacks

Because we previously had 6GB the memory of the virtual toolbox we had to increase the reserved RAM memory up to 9.2 GB. NR simulations operating at 100 MHz channel bandwidth with 100 UEs are exceptionally memory intensive. Running parallel batch execution scripts on standard development laptops (8–16 GB RAM) triggered OS Out-Of-Memory (OOM) process terminations.

Furthermore, because new ns-3 versions appear quite often with new features and properties we have decided to stick only to the ns-3 3.48 version. The project mandates using 5G-LENA v5.0 or later for the NR module in ns-3 to support handovers. Handover features in 5G-LENA are version-sensitive and require an exact corresponding ns-3 core release.

Moreover, the mandatory matrix consists of 55 total runs. High-density UE NR runs can exceed 30 minutes per execution. Our first run exceeded 1 hour.